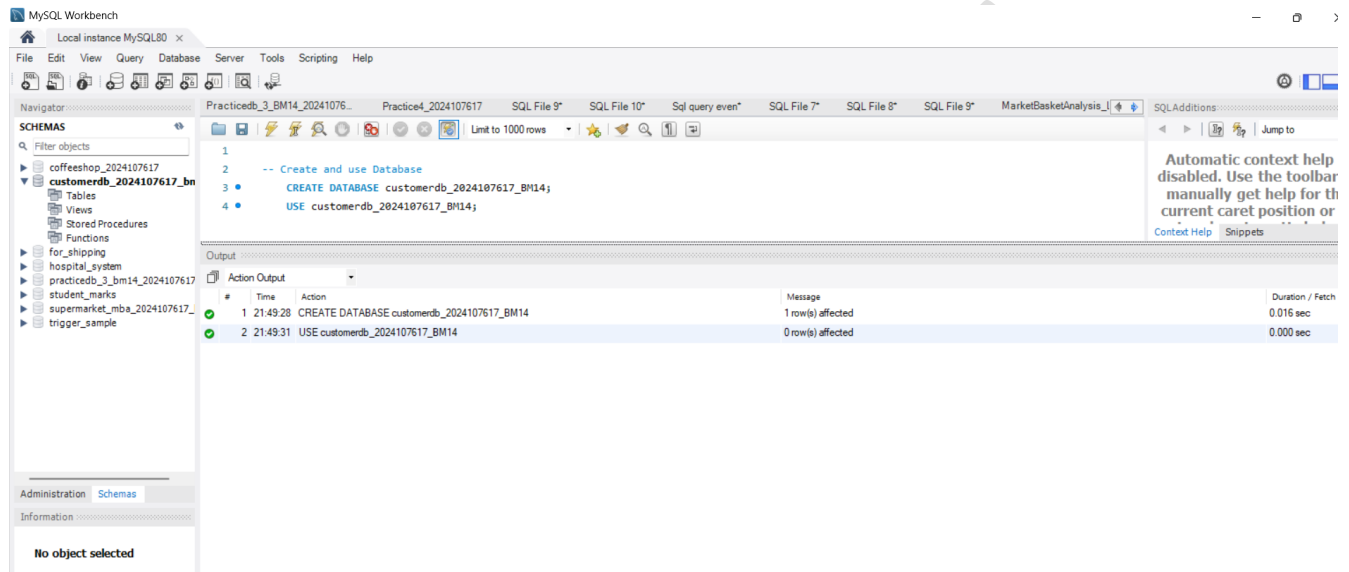


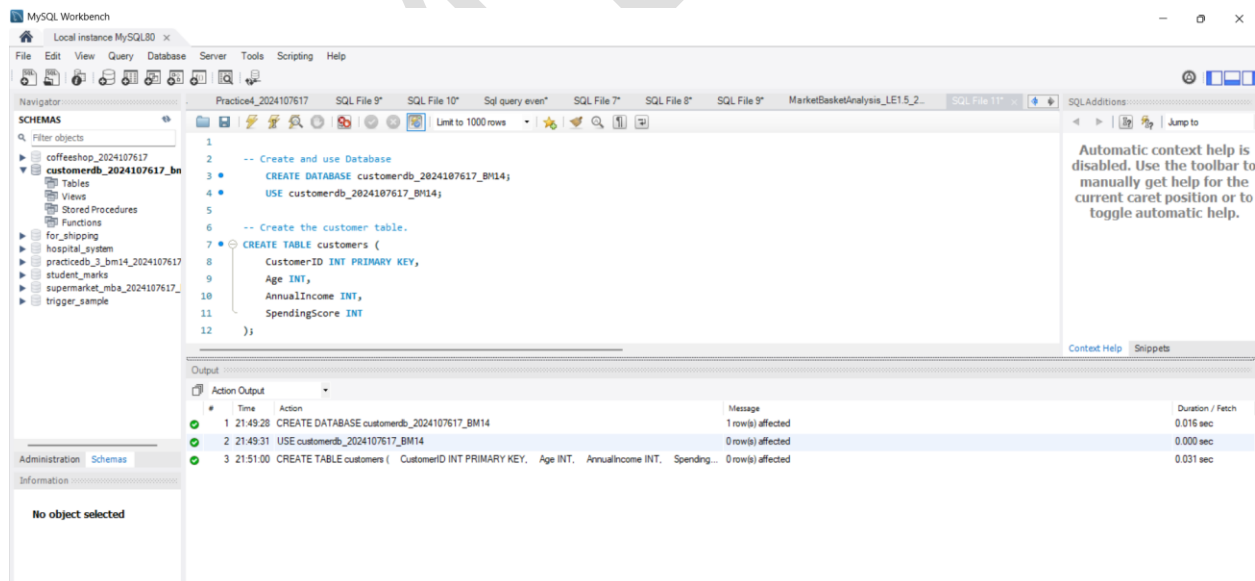
## SQL Results

## ACTIVITY 3 – IMPORTING DATA INTO MYSQL WORKBENCH

1. Create a database.
2. Use the database.

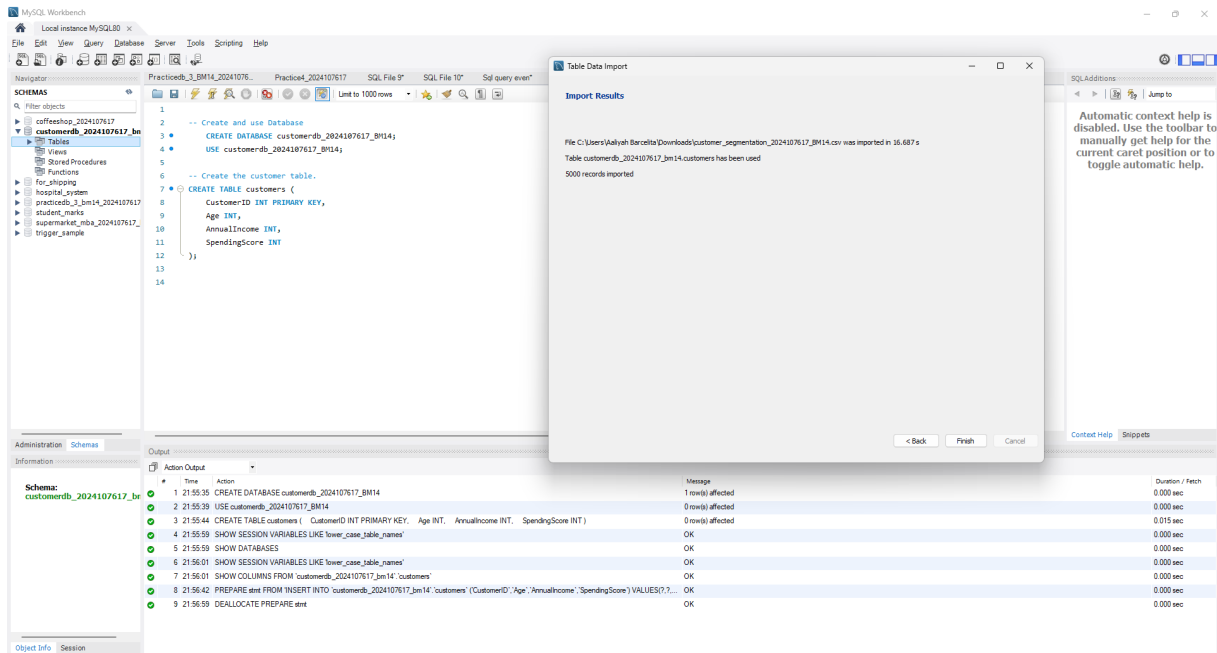


3. Create the customer table.



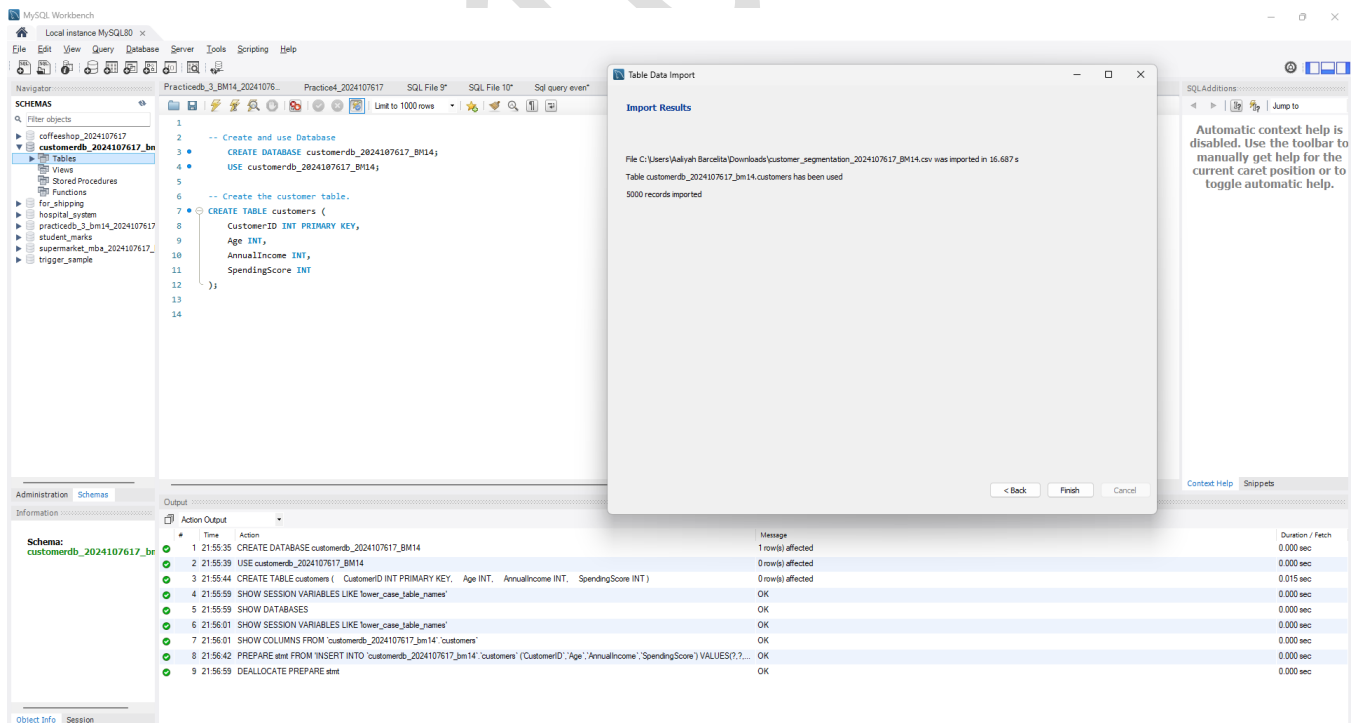
## LE1.6 DATA MINING 2 – CUSTOMER SEGMENTATION ANALYSIS

### 4. Import customer\_segmentation\_studentNumber\_Section.csv.



### 5. Verify successful import.

Show proof



## ACTIVITY 4 – EXPLORING CUSTOMER DATA

1. Run the following queries.
2. Find average income.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the 'customerdb\_2024107617' database selected. The main editor window contains the following SQL code:

```

11 SpendingScore INT
12 }
13 -- ACTIVITY 4 – EXPLORING CUSTOMER DATA
14 -- Find average income.
15 SELECT
16     AVG(AnnualIncome) AS Avg_Income
17 FROM customers;

```

The 'Result Grid' shows the output of the query:

| Avg_Income |
|------------|
| 67039.0128 |

The 'Output' pane shows the execution log with 10 rows of messages, including database creation, table creation, and session variables.

3. Find average spending score.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the 'customerdb\_2024107617' database selected. The main editor window contains the following SQL code:

```

17 FROM customers;
18 -- Find average spending score.
19 SELECT
20     AVG(SpendingScore) AS Avg_Spending_Score
21 FROM customers;

```

The 'Result Grid' shows the output of the query:

| Avg_Spending_Score |
|--------------------|
| 50.7952            |

The 'Output' pane shows the execution log with 11 rows of messages, including database creation, table creation, and session variables.

## LE1.6 DATA MINING 2 – CUSTOMER SEGMENTATION ANALYSIS

### 4. Find youngest customer.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
-- Find youngest customer.
SELECT
    MIN(Age) AS Youngest_Customer
FROM customers;
```

The left sidebar shows the database schema for 'customerdb\_2024107617'. The 'customers' table is selected, showing columns: CustomerID (int PK), Age (int), AnnualIncome (int), and SpendingScore (int). The bottom pane shows the 'Result Grid' with one row: Youngest\_Customer, 18. The 'Output' pane shows the execution log with 12 steps, including the final query execution which returned 1 row(s) in 0.015 seconds.

### 5. Find oldest customer.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
-- Find oldest customer.
SELECT
    MAX(Age) AS Oldest_Customer
FROM customers;
```

The left sidebar shows the database schema for 'customerdb\_2024107617'. The 'customers' table is selected, showing columns: CustomerID (int PK), Age (int), AnnualIncome (int), and SpendingScore (int). The bottom pane shows the 'Result Grid' with one row: Oldest\_Customer, 70. The 'Output' pane shows the execution log with 13 steps, including the final query execution which returned 1 row(s) in 0.000 seconds.

## LE1.6 DATA MINING 2 – CUSTOMER SEGMENTATION ANALYSIS

### ACTIVITY 5 – CUSTOMER SEGMENTATION USING SQL

1. Run the query.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
-- ACTIVITY 5 - CUSTOMER SEGMENTATION USING SQL
SELECT
  CustomerID, Age, AnnualIncome, SpendingScore,
  CASE
    WHEN AnnualIncome >= 80000 AND SpendingScore >= 70 THEN 'Premium Customers'
    WHEN AnnualIncome >= 80000 AND SpendingScore < 70 THEN 'Potential Customers'
    WHEN AnnualIncome < 80000 AND SpendingScore >= 70 THEN 'Frequent Shoppers'
    ELSE 'Budget Customers'
  END AS CustomerSegment
FROM customers;
```

The Results grid shows the following data:

| CustomerID | Age | AnnualIncome | SpendingScore | CustomerSegment     |
|------------|-----|--------------|---------------|---------------------|
| 1          | 64  | 90624        | 62            | Potential Customers |
| 2          | 34  | 39904        | 94            | Frequent Shoppers   |
| 3          | 18  | 101058       | 84            | Premium Customers   |
| 4          | 41  | 93936        | 1             | Potential Customers |
| 5          | 61  | 31042        | 78            | Frequent Shoppers   |
| 6          | 32  | 43439        | 81            | Frequent Shoppers   |
| 7          | 58  | 108319       | 60            | Potential Customers |

The Output tab shows the execution log:

| #  | Time     | Action                                                                                          | Message              | Duration / Fetch      |
|----|----------|-------------------------------------------------------------------------------------------------|----------------------|-----------------------|
| 14 | 22:31:01 | SELECT AVG(AnnualIncome) AS Avg_Income FROM customers LIMIT 0, 1000                             | 1 row(s) returned    | 0.016 sec / 0.000 sec |
| 15 | 22:31:12 | SELECT AVG(SpendingScore) AS Avg_Spending_Score FROM customers LIMIT 0, 1000                    | 1 row(s) returned    | 0.015 sec / 0.000 sec |
| 16 | 22:48:20 | SELECT CustomerID, Age, AnnualIncome, SpendingScore, CASE WHEN AnnualIncome >= 80000 AND Spe... | 1000 row(s) returned | 0.000 sec / 0.000 sec |
| 17 | 23:12:51 | SELECT CustomerID, Age, AnnualIncome, SpendingScore, CASE WHEN AnnualIncome >= 80000 AND Spe... | 5000 row(s) returned | 0.000 sec / 0.016 sec |

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
-- ACTIVITY 5 - CUSTOMER SEGMENTATION USING SQL
SELECT
  CustomerID, Age, AnnualIncome, SpendingScore,
  CASE
    WHEN AnnualIncome >= 80000 AND SpendingScore >= 70 THEN 'Premium Customers'
    WHEN AnnualIncome >= 80000 AND SpendingScore < 70 THEN 'Potential Customers'
    WHEN AnnualIncome < 80000 AND SpendingScore >= 70 THEN 'Frequent Shoppers'
    ELSE 'Budget Customers'
  END AS CustomerSegment
FROM customers;
```

The Results grid shows the following data:

| CustomerID | Age | AnnualIncome | SpendingScore | CustomerSegment     |
|------------|-----|--------------|---------------|---------------------|
| 4994       | 23  | 118997       | 50            | Potential Customers |
| 4995       | 27  | 92076        | 33            | Potential Customers |
| 4996       | 48  | 53752        | 73            | Frequent Shoppers   |
| 4997       | 56  | 83115        | 66            | Potential Customers |
| 4998       | 47  | 37040        | 75            | Frequent Shoppers   |
| 4999       | 28  | 112818       | 42            | Potential Customers |
| 5000       | 36  | 45538        | 37            | Budget Customers    |

The Output tab shows the execution log:

| #  | Time     | Action                                                                                          | Message              | Duration / Fetch      |
|----|----------|-------------------------------------------------------------------------------------------------|----------------------|-----------------------|
| 14 | 22:31:01 | SELECT AVG(AnnualIncome) AS Avg_Income FROM customers LIMIT 0, 1000                             | 1 row(s) returned    | 0.016 sec / 0.000 sec |
| 15 | 22:31:12 | SELECT AVG(SpendingScore) AS Avg_Spending_Score FROM customers LIMIT 0, 1000                    | 1 row(s) returned    | 0.015 sec / 0.000 sec |
| 16 | 22:48:20 | SELECT CustomerID, Age, AnnualIncome, SpendingScore, CASE WHEN AnnualIncome >= 80000 AND Spe... | 1000 row(s) returned | 0.000 sec / 0.000 sec |
| 17 | 23:12:51 | SELECT CustomerID, Age, AnnualIncome, SpendingScore, CASE WHEN AnnualIncome >= 80000 AND Spe... | 5000 row(s) returned | 0.000 sec / 0.016 sec |

## ACTIVITY 6 – SEGMENT SUMMARY REPORT

1. Run the query.

The screenshot shows the MySQL Workbench interface. The main editor displays a SQL query for customer segmentation. The query is as follows:

```

-- ACTIVITY 5 - CUSTOMER SEGMENTATION USING SQL
SELECT
  CustomerID, Age, AnnualIncome, SpendingScore,
  CASE
    WHEN AnnualIncome >= 80000 AND SpendingScore >= 70 THEN 'Premium Customers'
    WHEN AnnualIncome >= 80000 AND SpendingScore < 70 THEN 'Potential Customers'
    WHEN AnnualIncome < 80000 AND SpendingScore >= 70 THEN 'Frequent Shoppers'
    ELSE 'Budget Customers'
  END AS CustomerSegment
FROM customers;

-- ACTIVITY 6 - SEGMENT SUMMARY REPORT
SELECT

```

The Results Grid shows the output of the query:

| CustomerSegment     | TotalCustomers |
|---------------------|----------------|
| Potential Customers | 1274           |
| Frequent Shoppers   | 983            |
| Premium Customers   | 578            |
| Budget Customers    | 2165           |

The bottom panel shows the execution log with the following entries:

| #  | Time     | Action                                                                                          | Message              | Duration / Fetch      |
|----|----------|-------------------------------------------------------------------------------------------------|----------------------|-----------------------|
| 15 | 22:31:12 | SELECT AVG(SpendingScore) AS Avg_Spending_Score FROM customers LIMIT 0, 1000                    | 1 row(s) returned    | 0.015 sec / 0.000 sec |
| 16 | 22:48:20 | SELECT CustomerID, Age, AnnualIncome, SpendingScore, CASE WHEN AnnualIncome >= 80000 AND Spe... | 1000 row(s) returned | 0.000 sec / 0.000 sec |
| 17 | 23:12:51 | SELECT CustomerID, Age, AnnualIncome, SpendingScore, CASE WHEN AnnualIncome >= 80000 AND Spe... | 5000 row(s) returned | 0.000 sec / 0.016 sec |
| 18 | 23:18:54 | SELECT CASE WHEN AnnualIncome >= 80000 AND SpendingScore >= 70 THEN 'Premium Customers' WH...   | 4 row(s) returned    | 0.016 sec / 0.000 sec |